RESTAURANT GROWTH POTENTIAL MODELLING

& STRATEGIC CLASSIFICATION SYSTEM

An End-to-End Data-Driven Analysis for Expansion Readiness

Name	Hari Ganesh S A
UNID	UMID0801267752
Role	Data Science Intern
Project Email	hs2110@srmist.edu.in
Gmail (Payment)	hariganeshsa@gmail.com
Submission Date	Before Apr 15th 2026

1. Introduction

Restaurant expansion is one of the most capital-intensive and risk-prone strategic decisions in the food service industry. Traditional decision-making often relies on surface-level indicators such as revenue growth or order volume, but these approaches frequently fail to account for cost pressure, channel dependency, scalability constraints, and structural inefficiencies. This often leads to businesses expanding high-volume restaurants that later struggle to sustain profitability in the long run.

This project is designed to move beyond simplistic growth metrics and build a holistic, data-driven framework for evaluating restaurant expansion readiness. By integrating exploratory data analysis (EDA), advanced feature engineering, machine learning clustering algorithms, and a bespoke Growth Potential Index (GPI), the analysis provides actionable insights into which restaurants are truly positioned for sustainable expansion and which require structural optimization before scaling.

2. Business Problem Statement

The central business problem addressed in this project is:

> ***How can restaurants be objectively classified based on their true expansion readiness — rather than superficial growth indicators — while identifying the operational levers that most influence sustainable scaling?***

2.1 Key Challenges Addressed

- High order volume masking low profitability — top-line growth does not always reflect bottom-line health
- Heavy dependence on third-party aggregators (Uber Eats, DoorDash) compressing net margins
- Rising cost pressure during aggressive growth phases — COGS and OPEX spiraling together
- Lack of a unified metric to assess expansion readiness across diverse restaurant formats
- Absence of strategic segmentation for portfolio-level decision-making

2.2 Three Core Analytical Pillars

- Holistic Growth-Readiness Assessment — portfolio, geographic, and segment-level evaluation
- Data-Driven Restaurant Classification — machine learning-powered cluster segmentation
- Identification of Growth Levers — operational factors that drive or constrain expansion

3. Data Understanding & Preparation

3.1 Dataset Overview

The dataset comprises restaurant-level operational, financial, and channel-specific data from SkyCity Auckland Restaurants & Bars. The final working dataset contained 1,696 restaurant records with 34 columns after feature engineering — enabling rich, multi-dimensional analysis.

Feature Category	Key Features	Details
Channel Orders	InStore, UberEats, DoorDash, SelfDelivery	Order counts per channel
Channel Revenue	UberEatsRevenue, DoorDashRevenue, etc.	Revenue per delivery channel
Cost Components	COGS Rate, OPEX Rate, Commission Rate	Percentage-based cost ratios
Growth Metrics	GrowthFactor, MonthlyOrders, AOV	Expansion readiness signals
Business Attributes	Segment, CuisineType	5 cuisine types, 4 segments
Geographic	Subregion (5 zones)	Auckland geographic breakdown
Delivery Metrics	DeliveryRadiusKM, DeliveryCostPerOrder	Range 3–18 km
Profit Metrics	InStore, UberEats, DoorDash, SD NetProfit	Per-channel net profit

3.2 Dataset Statistics

- Total Records: 1,696 restaurant observations
- Total Columns (post-engineering): 34 features
- Monthly Orders Range: 441 to 2,337 orders
- Average Order Value (AOV) Range: $29.79 to $47.23
- GrowthFactor Range: 1.00 to 1.05 (mature, stable market)
- COGS Rate Range: 20% to 40% — significant variability
- OPEX Rate Range: 24.5% to 54.9% — highest source of cost dispersion
- Commission Rate Range: 27% to 33%
- Delivery Radius: 3 km to 18 km across subregions

3.3 Data Cleaning Process

The cleaning process focused on ensuring business realism and analytical consistency across all downstream analyses:

- Missing values in operational metrics were validated against business logic and corrected
- Channel-wise order counts were cross-checked against total MonthlyOrders figures for consistency
- Derived profit metrics were recalculated where arithmetic inconsistencies were detected
- Revenue and cost ratios were validated to ensure values stayed within plausible real-world ranges
- Restaurant IDs were verified to ensure uniqueness and no duplicate records existed
- Categorical fields (CuisineType, Segment, Subregion) were standardized for uniform encoding

4. Outlier Identification & Treatment

4.1 Outlier Identification Methodology

Outliers were detected across all numerical columns using the Interquartile Range (IQR) method — the industry standard for non-parametric outlier detection. For each numerical column, the following bounds were computed:

Statistic	Formula	Purpose
Q1 (25th Percentile)	raw_df[col].quantile(0.25)	Lower quartile boundary
Q3 (75th Percentile)	raw_df[col].quantile(0.75)	Upper quartile boundary
IQR	Q3 - Q1	Spread of middle 50% of data
Lower Limit	Q1 - 1.5 * IQR	Minimum acceptable threshold
Upper Limit	Q3 + 1.5 * IQR	Maximum acceptable threshold

4.2 Outlier Treatment Strategy: IQR Capping (Winsorization)

Rather than removing outlier records (which would reduce the dataset), a capping/winsorization strategy was applied. Values below the lower limit were capped to the lower limit, and values above the upper limit were capped to the upper limit using the pandas mask() function:

```
raw_df[col] = raw_df[col].mask(raw_df[col] < lower_limit, lower_limit)
raw_df[col] = raw_df[col].mask(raw_df[col] > upper_limit, upper_limit)
```

4.3 Rationale for Capping Over Deletion

- Preserves all 1,696 records, maintaining statistical power for clustering
- Extreme values in hospitality data often represent genuine peak-season or special event performance
- Capping retains the directional signal of high-performance restaurants while preventing algorithm dominance
- Post-treatment verification confirmed zero remaining outliers across all numerical columns
- Prevents StandardScaler from being skewed by extreme magnitude features during clustering

5. Exploratory Data Analysis (EDA)

5.1 Holistic Growth-Readiness Assessment

EDA was conducted across three analytical layers — portfolio, geographic, and segment-level — to establish a comprehensive picture of growth readiness before any machine learning was applied.

Portfolio-Level Findings

- Stable Performers dominate the portfolio at 44.4% of all restaurants, yet generate only approximately 52% of the order volume produced by High-Growth restaurants — indicating demand concentration risk
- High-Growth restaurants (19.0% of portfolio) produce the highest average monthly orders at 1,531 but show a negative average GPI of -0.08, revealing that growth is driven by cost-heavy expansion rather than structural efficiency
- Aggregator-Dependent Low-Margin restaurants (36.6%) exhibit the highest average GPI of +0.31, proving that platform dependence does not automatically negate growth potential when cost structures are well-managed
- The top 10 restaurants alone generate between 1,550 and 1,770 monthly orders each, confirming that demand is highly skewed toward an elite sub-group — reinforcing the need for targeted rather than broad-based scaling

Geographic-Level Findings

- South Auckland leads geographically with the highest average GPI of +0.34 — driven by lower real estate costs, favorable cost structures, and strong community demand
- CBD records the lowest average GPI of -0.58 despite comparable or higher order volumes — confirming that cost structures and regional economics matter more than raw demand
- West Auckland and North Shore show moderate GPI values, representing stable secondary markets suitable for incremental expansion
- Geographic analysis strongly suggests expansion strategy should prioritize cost-efficient suburban zones over premium urban locations

Segment-Level Findings

- Quick Service Restaurants (QSRs) form the largest segment at 32% of the portfolio but are not the most growth-ready
- Ghost Kitchens (11.6% of portfolio) deliver the highest average GPI of +0.12, validating that asset-light, delivery-first models outperform size-driven models in growth readiness
- Cafes and Full-Service Restaurants show higher cost pressure due to staffing and real estate requirements
- The GPI hierarchy by segment is: Ghost Kitchen > QSR > Cafe > Full Service Restaurant

5.2 Data-Driven Restaurant Classification Analysis

Channel Order Distribution

- Uber Eats contributes the largest share of total orders across the portfolio at approximately 800,000 orders — more than any single other channel
- In-store orders rank second but show the highest per-order revenue yield, as no commission is deducted

- DoorDash and Self-Delivery collectively contribute a smaller share of volume but generate proportionally better unit economics per order
- The aggregate dependence on Uber Eats creates systemic platform concentration risk across the entire portfolio

Revenue vs. Order Volume Analysis

- Revenue does not scale proportionally with orders across channels — a critical finding
- Self-Delivery generates comparable revenue to DoorDash with significantly fewer orders, indicating superior per-order revenue capture of approximately $38.50 vs. $37.30
- Uber Eats leads in total revenue (~$18K average per restaurant) due to volume, not per-order economics
- Commission rates of 27–33% compress Uber Eats net profit significantly, reducing effective margin

Cuisine-Level Growth Analysis

- Pizza and Burgers lead all cuisine types in average monthly orders — but only Pizza shows a consistently positive GPI
- Burgers generate high volume but exhibit weak or negative GPI due to high COGS and commission pressure
- Indian cuisine shows high order volumes with negative GPI — suggesting structural cost imbalance
- Thai and Kebabs/Mediterranean cuisines exhibit the most negative GPI values, flagging significant expansion risk
- Only Pizza has demonstrated consistent positive GPI alongside high demand — making it the most expansion-ready cuisine category

5.3 Growth Levers Impacting Expansion Readiness

Four primary growth levers were evaluated against cluster labels to identify what truly drives or constrains expansion readiness:

Growth Lever	High-Growth	Stable Performers	Dependent Low Margin
Aggregator Dependency	~55% (lowest)	~62%	~70% (highest)
Monthly Orders	~1,531 (highest)	~804	~1,180
Average Order Value	$38–$39	$38–$39	$38–$39
COGS + OPEX Rate	~88% (critical)	~72%	~76%

- Lever 1 — Aggregator Dependency: High-Growth restaurants process ~55% of orders via aggregators — the lowest among all clusters. Better channel control is a critical pre-condition for scalable growth

- Lever 2 — Demand Strength: The High-Growth cluster leads with ~1,531 average monthly orders — nearly 90% higher than Stable Performers at 804. Demand remains the strongest positive expansion signal
- Lever 3 — AOV (Average Order Value): Flat at $38–$39 across all three clusters, contributing less than 3% variation. AOV is definitively not a differentiating growth lever
- Lever 4 — Cost Pressure: COGS + OPEX peaks at approximately 88% in High-Growth restaurants. Despite strong demand, cost inefficiency is the primary expansion risk — making cost control the most critical intervention lever

6. Feature Engineering

6.1 Purpose of Feature Engineering

Raw operational and financial data alone cannot capture the nuanced dynamics of expansion readiness. Feature engineering was applied to derive composite metrics that better represent the business problem from a multi-dimensional perspective.

6.2 Key Engineered Features

- Aggregator Dependency Ratio: (UberEatsOrders + DoorDashOrders) / MonthlyOrders — measures the proportion of demand dependent on third-party platforms
- Channel Contribution Percentages: InStoreShare, UE_share, DD_share, SD_share — percentage of total orders from each channel
- Net Profit per Channel: InStoreNetProfit, UberEatsNetProfit, DoorDashNetProfit, SelfDeliveryNetProfit — channel-level profit after costs and commissions
- SD Total Delivery Cost: SelfDeliveryOrders * DeliveryCostPerOrder — operational cost of managing self-delivery logistics
- Cost Pressure Composite: COGSRate + OPEXRate — combined structural cost burden as a percentage of revenue
- Growth Potential Index (GPI): Multi-dimensional composite score combining demand, revenue quality, cost resilience, and channel scalability (detailed in Section 7)

6.3 Feature Engineering Impact

- The raw dataset of approximately 20 base features expanded to 301+ columns post one-hot encoding for ML modeling
- Engineered ratio features reduce scale sensitivity and enable better cross-restaurant comparisons
- Channel share features normalize order volumes, making small and large restaurants directly comparable
- Net profit features expose the true economics hidden within gross revenue figures

7. Feature Selection

7.1 GPI Feature Set

Feature selection for the Growth Potential Index was driven by business logic and domain knowledge rather than purely statistical criteria. Seven key features were selected to represent the four dimensions of growth readiness:

Feature	GPI Weight	Direction	Business Rationale
GrowthFactor	+0.30	Positive	Primary demand expansion signal
MonthlyOrders	+0.25	Positive	Volume strength and market reach
AOV	+0.15	Positive	Revenue quality per transaction
DeliveryRadiusKM	+0.05	Positive	Geographic scalability indicator
COGSRate	-0.10	Negative	Direct cost pressure on margin
OPEXRate	-0.10	Negative	Operational overhead burden
CommissionRate	-0.05	Negative	Platform fee drag on net margin

7.2 Clustering Feature Set

For K-Means and Hierarchical clustering, all 301+ encoded and scaled features were used to capture the full dimensionality of restaurant behavior. PCA was subsequently applied for visualization purposes only — the clustering itself was performed in high-dimensional scaled feature space.

8. Encoding & Scaling

8.1 One-Hot Encoding

Categorical variables were encoded using Scikit-learn's OneHotEncoder with drop='first' to prevent multicollinearity. Three categorical columns were transformed:

- Segment — 4 categories (QSR, Cafe, Full Service, Ghost Kitchen) encoded to 3 binary columns
- CuisineType — 6 categories (Burgers, Pizza, Indian, Thai, Kebabs/Mediterranean, Other) encoded to 5 binary columns
- Subregion — 5 geographic zones (CBD, South Auckland, North Shore, West Auckland, East Auckland) encoded to 4 binary columns

Important: One-hot encoded features were used exclusively for machine learning models. All business interpretation, EDA visualizations, and GPI calculations used the original categorical labels for readability and interpretability.

8.2 Standard Scaling

All numerical features were normalized using StandardScaler (zero mean, unit variance) before clustering. This ensures:

- No single high-magnitude feature (such as MonthlyOrders or Revenue) dominates distance calculations in K-Means
- Fair Euclidean distance computation across features measured in different units (orders, dollars, percentages, kilometres)
- Improved convergence stability in the K-Means optimization algorithm
- PCA components are not disproportionately influenced by high-variance raw features

Critical Note: Scaled and encoded data were strictly used only for modeling purposes. All business interpretation, cluster profiling, and strategic recommendations were derived from raw, unscaled feature values to maintain real-world interpretability.

9. Growth Potential Index (GPI)

9.1 Purpose & Design

The Growth Potential Index (GPI) is the central innovation of this project. It was designed to provide a single, composite, interpretable score that captures multi-dimensional expansion readiness — going beyond simple revenue or order volume metrics.

9.2 GPI Formula

```
GPI = 0.30 * GrowthFactor + 0.25 * MonthlyOrders + 0.15 * AOV + 0.05 *
DeliveryRadiusKM  -  0.10 * COGSRate  -  0.10 * OPEXRate  -  0.05 *
CommissionRate
```

All input features are first standardized using StandardScaler before the GPI formula is applied, ensuring each component contributes proportionally according to its assigned weight rather than its raw magnitude.

9.3 GPI Interpretation

- GPI > 0.5: High expansion readiness — strong demand with controlled costs
- GPI 0.1 to 0.5: Moderate readiness — suitable for conditional expansion with monitoring
- GPI -0.1 to 0.1: Neutral zone — optimization required before expansion decisions
- GPI < -0.1: Low readiness — structural issues in cost or channel mix must be resolved first

9.4 Cluster-Wise GPI Summary

Cluster Label	Mean GPI	Min GPI	Max GPI
Dependent Low Margin	+0.31	-0.21	+1.30
Stable Performers	+0.09	-0.98	+0.89
High-Growth / High-Risk	-0.08	-1.37	+0.82

10. Machine Learning Models

10.1 Dimensionality Reduction via PCA

Principal Component Analysis (PCA) was applied with a variance retention threshold of 0.85 (85%), reducing the 301+ feature space to a significantly smaller set of principal components. This enabled computationally efficient clustering while retaining the majority of informational variance.

- Original features: 301+ columns post one-hot encoding
- PCA components retained (for visualization): 2 components
- PCA purpose: Strictly for 2D scatter plot visualization — not used as input to clustering
- Clustering algorithm used the full StandardScaler-transformed feature space

10.2 K-Means Clustering

Optimal K Selection

The optimal number of clusters was determined using a combination of the Elbow Method (Inertia minimization) and Silhouette Score maximization, evaluated across K values from 2 to 10:

- Elbow Method: Inertia showed a clear inflection point at K=3, after which additional clusters yielded diminishing inertia reduction
- Silhouette Score: K=3 produced the highest average silhouette score, confirming well-separated, internally cohesive clusters
- Business validation: Three clusters aligned naturally with three distinct operational archetypes in the hospitality domain

K-Means Results (K=3)

- Final model: KMeans(n_clusters=3, random_state=42) fit on StandardScaler-transformed data
- Cluster 0 — High-Growth / High-Risk Expanders: 19% of restaurants with highest monthly orders (~1,531 avg) but highest cost pressure (COGS+OPEX ~88%)
- Cluster 1 — Stable Core Performers: 44.4% of restaurants — balanced demand, controlled costs, predictable performance
- Cluster 2 — Aggregator-Dependent / Channel-Driven: 36.6% of restaurants — heavy platform reliance but often higher GPI due to operational efficiency

PCA Scatter Plot Observations

- Clear visual separation into three dense regions is visible in the 2D PCA projection
- Cluster 0 (High-Growth) appears as a tight cluster in the high-variance region of PC1
- Cluster 1 (Stable) forms a diffuse, spread-out region indicating heterogeneity within the stable category
- Cluster 2 (Aggregator-Dependent) shows a distinct positioning with moderate PC1 and varying PC2 scores
- Partial overlap at cluster boundaries is expected and represents restaurants in transitional states

10.3 Hierarchical Clustering

Agglomerative Hierarchical Clustering with Ward linkage was applied as a validation and comparison model. The Ward linkage minimizes total within-cluster variance at each merge step — producing compact, equal-variance clusters.

Dendrogram Analysis

- The dendrogram reveals a clear top-level split into exactly 3 major branches at a distance threshold of approximately 15–18 units
- Sub-branches within each major cluster reveal micro-level similarity structures — useful for future granular segmentation
- The three-cluster structure is confirmed robustly — cutting the dendrogram at any reasonable threshold between 10 and 20 consistently produces 3 groups
- Cluster sizes from hierarchical clustering closely match K-Means cluster sizes — validating the segmentation as structurally sound

Hierarchical Clustering Observations

- High-Growth branch: characterized by tight sub-clustering around high-order, high-cost restaurants
- Stable Performers branch: the broadest branch with the most internal sub-structure, reflecting diversity within this large cluster
- Aggregator-Dependent branch: moderately tight, suggesting a more homogeneous group driven by channel composition patterns

10.4 DBSCAN (Density-Based Clustering)

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was applied as a robustness check to determine whether restaurant data exhibits natural density-based clustering structures.

K-Distance Plot Analysis

- The K-Distance plot (K=5) was used to determine the optimal epsilon (eps) parameter
- The plot shows a gradual, continuous curve rather than a sharp elbow — indicating no strong density discontinuities in the data

- Optimal eps was set at 1.2 based on the approximate inflection point of the K-distance graph

DBSCAN Results

- Almost all data points (>95%) were labeled as noise (cluster label = -1)
- No meaningful dense clusters were detected at any reasonable eps/min_samples combination
- Conclusion: Restaurant operational data follows a continuous distribution without natural density gaps — making DBSCAN fundamentally unsuitable for this segmentation problem
- DBSCAN served as an important robustness check confirming that K-Means and Hierarchical approaches are the appropriate methods

11. Model Performance & Comparison

11.1 Comparative Model Evaluation

Criterion	K-Means	Hierarchical	DBSCAN	Winner
Cluster Separation	Clear (3 groups)	Clear (3 branches)	Fails (all noise)	K-Means / HC
Silhouette Score	Best at K=3	Comparable	N/A	K-Means
Scalability	High	Moderate	Low	K-Means
Business Interpretability	High	High	None	Both
Handles Density Variation	Limited	Limited	Designed for it	DBSCAN (unsuitable here)
Cluster Count Control	Explicit (K=3)	Cut dendrogram	None	K-Means
Deployment Suitability	High	Medium	None	K-Means

11.2 Final Model Selection

K-Means Clustering (K=3) was selected as the primary deployment model for the following reasons:

- Produces the best-defined, most interpretable three-cluster structure validated by both Elbow Method and Silhouette Score
- Hierarchical clustering serves as an independent validation mechanism — and both methods agree on the three-cluster structure
- DBSCAN was disqualified due to near-complete noise labeling — confirming the data's continuous, non-density-based distribution
- K-Means is computationally efficient and operationally scalable for quarterly portfolio reviews with updated restaurant data

- The cluster labels produced by K-Means map cleanly onto recognizable business archetypes, enabling straightforward communication to business stakeholders

12. Clustering Analysis & Observations

12.1 Cluster Profiling

After K-Means fit, each cluster was profiled using raw (unscaled) feature means to derive business-interpretable characteristics:

Feature	High-Growth (Cluster 0)	Stable Performers (Cluster 1)	Dependent Low-Margin (Cluster 2)
Avg Monthly Orders	~1,531	~804	~1,180
Avg GrowthFactor	~1.045	~1.015	~1.038
Avg AOV ($)	~$38.6	~$38.5	~$38.5
COGS Rate	~34%	~27%	~26%
OPEX Rate	~54%	~40%	~44%
Aggregator Ratio	~55%	~62%	~70%
Avg GPI	-0.08	+0.09	+0.31
Delivery Radius (km)	~12 km	~9 km	~11 km

12.2 Cluster 0: High-Growth / High-Risk Expanders

- Strongest demand signal in the portfolio with average monthly orders of approximately 1,531 — 90% higher than Stable Performers
- Widest delivery radius (average ~12 km), indicating strong geographic reach and willingness to serve dispersed customer bases
- Despite demand strength, this cluster carries the most dangerous cost profile: combined COGS + OPEX averaging approximately 88%
- Negative average GPI of -0.08 reveals that volume growth is not translating into structural expansion readiness
- Lower aggregator dependency (~55%) suggests some degree of channel diversification, which is a positive structural indicator
- Strategic Implication: These restaurants have the demand foundation for expansion but MUST implement strict cost ceilings and margin safeguards before scaling. Scaling without cost control will amplify losses.

12.3 Cluster 1: Stable Core Performers

- Forms the largest cluster at 44.4% of the portfolio — representing the backbone of the restaurant network
- Moderate monthly orders averaging approximately 804 — consistent, predictable, but not explosive
- Best cost structure in the portfolio with COGS ~27% and OPEX ~40%, yielding the lowest combined cost pressure
- Highest aggregator dependency at ~62% represents a concentration risk that could erode margins if platform fees increase
- Positive average GPI of +0.09 confirms structural health, though not positioned for aggressive expansion
- Strategic Implication: Optimize for efficiency improvements and selective expansion. Prioritize reducing aggregator dependency and growing self-delivery or in-store channels to improve margin quality.

12.4 Cluster 2: Aggregator-Dependent / Channel-Driven Growth Candidates

- The counterintuitive star performer with the highest average GPI of +0.31 — despite its 'dependent' label
- Moderate-to-high monthly orders averaging approximately 1,180 with strong GrowthFactor (~1.038)
- Highest aggregator dependency at ~70%, yet efficiently monetizes this platform exposure with controlled COGS (~26%)
- The gap between high platform dependence and strong GPI reveals that operational efficiency can compensate for distribution costs
- Represents 36.6% of the portfolio — a significant and strategically important segment
- Strategic Implication: Scale selectively with focus on channel rebalancing. Reducing aggregator share to 55–60% while maintaining current operational efficiency would dramatically improve net margin and GPI further.

12.5 Cross-Cluster Strategic Observations

- The GPI ranking inverts the intuitive order: Dependent Low-Margin > Stable Performers > High-Growth. This counter-intuitive finding is the central strategic insight of the entire project.
- High demand does not guarantee expansion readiness — cost structure and channel efficiency are more important determinants of GPI
- The healthiest expansion candidates are not in the highest-order cluster, but in the most operationally efficient cluster
- Ghost Kitchens and QSRs in Cluster 2 dominate the high-GPI rankings, validating the asset-light business model thesis
- South Auckland restaurants appear disproportionately in Cluster 2, reinforcing the geographic advantage of cost-efficient suburban markets

13. Strategic Recommendations

13.1 Expansion Decision Framework

- Use GPI as the primary gating metric for expansion approval — not monthly order volume or revenue alone
- Set a minimum GPI threshold of +0.15 for expansion approval, with a target threshold of +0.30 for priority expansion
- Scale only high-GPI restaurants (predominantly from Cluster 2) as the first expansion priority
- Impose strict cost ceilings: COGS should not exceed 32% and OPEX should not exceed 48% before expansion approval
- Prioritize South Auckland and similar suburban markets over CBD-style cost-heavy locations

13.2 Operational Recommendations

- Reduce aggregator dependency below 55% across all clusters through investment in owned channels (app ordering, self-delivery)
- Invest in Ghost Kitchen scalability as the highest-GPI, lowest-capex expansion format
- Rebalance channel mix for low-margin cuisines (Burgers, Indian) before approving expansion
- Diversify growth demand away from the top-10 restaurant concentration to reduce portfolio fragility
- Implement quarterly GPI-based portfolio reviews using the K-Means clustering framework as an ongoing decision tool

14. Conclusion

This project demonstrates that sustainable restaurant expansion is not driven by demand alone, but by a complex interaction of channel control, cost efficiency, regional economics, and business model design. By combining advanced EDA, machine learning clustering, and the bespoke Growth Potential Index, the framework provides a robust, repeatable, and executive-ready decision system for expansion readiness assessment.

The most striking finding is the counter-intuitive GPI ranking: Aggregator-Dependent Low-Margin restaurants (+0.31 avg GPI) outperform High-Growth restaurants (-0.08 avg GPI) in expansion readiness. This proves that structural efficiency — not raw volume — is the true determinant of sustainable scaling capability.

Going forward, expansion decisions should be gated by GPI rather than volume metrics, focused on cost-efficient regions and scalable business models. Restaurants must not be expanded solely on demand spikes or aggregator-driven growth without margin safeguards. This framework provides a data-driven, repeatable methodology to decide where to scale, where to optimize, and where to restructure before committing expansion capital.

Final Answer: Which restaurants are structurally positioned for sustainable growth?

Restaurants combining strong consistent demand, controlled cost structures (COGS < 32%, OPEX < 48%), balanced channel mix (aggregator dependency < 60%), and positive GPI scores are structurally positioned for sustainable growth. Based on this project's analysis, the Aggregator-Dependent Low-Margin cluster — particularly Ghost Kitchens and QSRs in South Auckland — represents the most expansion-ready segment in the current portfolio.

Submitted by: Hari Ganesh S A | UMID0801267752 | Data Science Intern

hs2110@srmist.edu.in | hariganeshsa@gmail.com